

Chao Xu

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Professor of algorithms with a production-engineering track: ~30 papers in combinatorial optimization (SODA, SICOMP, Math. Programming, AAAI) on one side, engineering years at Google, Yahoo Research, Grab AI, and Voleon on the other.

I work toward AI agents that reason reliably — that search, plan, verify, and discover mathematical arguments rather than reproduce benchmark patterns. I contribute by designing hard reasoning tasks, building evaluation environments where correctness is mechanically checkable, and bringing proof and optimization structure into agentic scientific workflows.

Self-Directed AI Systems

- **Autonomous proof-search system** (*public*) — an agent harness (**coverify**) that attacks open problems from my public [research catalog](#) over multi-day runs: durable state files, subagent fan-outs, sandboxed execution, scheduled verification passes. Candidate proofs survive only independent review; results land back in a Forgejo-native math knowledge base (**cosheaf**), edited with a math-aware Markdown editor built on CodeMirror 6 (**coflat**). Campaigns have closed multiple open problems in combinatorial optimization; two are public in the catalog's solved archive, the rest are being written up for publication.

Experience

- **CEO** — **Max Flow** (2022 – present). Technology-first consumer-electronics distributor built on retail arbitrage (mostly laptops): automated sourcing below promotional rates plus warehouse-integration tooling, run by an operations-research/CS team; ~\$20M annual revenue.
- **Associate Professor** — **UESTC**, Chengdu (2023 – present; Asst. Prof. 2022). Lead a research group on combinatorial optimization and applied algorithms; advise students through to publication (ISAAC, ECAI, IJCAI, AAAI with student first authors); 10+ papers in this period.
- **Visiting Researcher** — **Tepper, Carnegie Mellon University** (Jan – Jun 2023). Hosted by Benjamin Moseley.
- **Software Engineer** — **Voleon**, Berkeley (2020 – 2021). Research Engineering at a quantitative investment firm: built and operated production research/backtesting infrastructure used directly by ML researchers — clean training/experiment abstractions, reproducible runs, fast iteration.
- **Senior Software Engineer** — **Grab AI Accelerator**, Bellevue (2019 – 2020). Productionized ML algorithm work for Southeast Asia's largest ride-hailing/fintech platform.
- **Research Scientist** — **Yahoo Research**, New York (2018 – 2019). Scalable Machine Learning Group; algorithms for large-scale ML systems.
- **Software Engineer** — **Google**, Mountain View (2013). Google Analytics backend; algorithmic improvements to the optimization package.

Selected Publications

Full list at chaoxu.prof. Highlights: **A Faster Pseudopolynomial Time Algorithm for Subset Sum** (SODA 2017; SICOMP 2019), **Hypergraph k-Cut in Randomized Polynomial Time** (SODA 2018; Math. Programming 2019), **Detecting Weakly Simple Polygons** (SODA 2015), **Linear TreeShap** (NeurIPS 2022 Oral), **Binary Split Categorical Feature with MAE Criteria in CART** (AAAI 2026).

Education

- **PhD, Computer Science** — **University of Illinois at Urbana-Champaign** (2013 – 2018). Advisors: Karthekeyan Chandrasekaran and Chandra Chekuri. Dissertation in combinatorial optimization (graph cuts, submodular functions).
- **BS, Mathematics & Applied Math/Statistics, minor in CS** — **Stony Brook University** (2009 – 2013).

Skills & Competitions

Python, Haskell, APL, SQL, Java; TypeScript/Bun for agent tooling, including MCP servers and CLI harnesses. Daily Codex and Claude Code user for shipped projects; gives invited talks on agent-harness and context-engineering design for research work. ACM-ICPC Greater NY Regional 2nd place (2012); Putnam top 400 (2010).